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FLUID-STRUCTURE INTERACTION EFFECTS ON PAYLOAD DEFLECTION AND TIP-OVER STABILITY IN MOBILE CRANES UNDER WIND LOADING

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Abstract

Mobile crane stability under wind loading is commonly evaluated using quasi-static load charts and equivalent-static wind formulations that assume rigid geometry and constant forces. These approaches neglect deformation-dependent effects associated with suspended payloads and flexible structures that can significantly influence tip-over conditions. This paper presents a quasi-static numerical investigation of wind-induced stability in a telescopic mobile crane using a coupled computational fluid dynamics (CFD) and one-way fluid–structure interaction (FSI) framework. Steady CFD simulations are used to predict aerodynamic forces and moments under varying wind directions, velocities, and payload geometries, which are then applied to a structural model to quantify wind-induced payload deflection. Results indicate that crosswind loading produces substantially higher overturning moments than along-wind conditions, while payload geometry strongly influences aerodynamic forces, with bluff cubic payloads generating larger moments than spherical payloads of comparable mass. The results further show that wind-induced deformation increases the effective moment arm of aerodynamic forces, introducing additional overturning effects not captured by rigid-body or standards-based methods. Overall, the findings demonstrate that, under mean wind loading, deformation-dependent effects play a significant role in crane stability and should be considered in stability assessment.

Keywords: Mobile crane dynamics, Wind load stability, CFD-FSI coupling, Payload swing, Overturning moment, Tip-over

1. Introduction

Mobile cranes play a critical role in modern construction, infrastructure development, and heavy industrial operations. Their ability to lift and precisely position large structural components, mechanical equipment, and prefabricated assemblies makes them indispensable in dense urban construction sites, ports, power plants, and refineries. Mobile telescopic boom cranes, illustrated

FIGURE 1: Telescopic boom mobile crane.

in Figure 1, enable operation in environments where fixed lifting systems are impractical or infeasible. At the same time, these favorable characteristics introduce significant challenges for stability, as crane configurations vary continuously during operation and often involve long lever arms, elevated centers of mass, and suspended payloads subject to environmental disturbances [1]. Crane tip-over accidents represent one of the most severe failure modes in lifting operations and are frequently associated with extensive property damage, schedule disruption, and catastrophic injuries. While overturning is often described as a sudden loss of stability, recent analysis of real-world accidents indicate that the tip-over process for tall mobile cranes evolves over tens of seconds, depending on boom length and configuration. Video-based accident studies by Stone et al. show that mobile crane tip-over durations commonly range from 5s to over 30s, with longer booms exhibiting slower angular acceleration during the early stages of

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instability [2]. This transitional phase is characterized by gradual unloading of the outriggers and increasing rotational motion about the tipping edge. In many documented cases, operators report perceiving the crane "getting light," providing a limited, but meaningful, opportunity for corrective actions such as raising the boom, or lowering the payload before full overturning occurs.

Investigations by the Occupational Safety and Health Administration (OSHA) and the National Institute for Occupational Safety and Health (NIOSH) indicate that mobile crane overturning incidents often occur during routine lifting operations rather than extreme overload scenarios, highlighting the narrow margin between stable and unstable configurations in practice [3, 4].

The stability of a mobile crane is governed by the interaction of several factors, including payload mass, boom length and inclination, counterweight configuration, ground support conditions, and structural response under applied loads. Environmental loading, particularly wind, introduces additional destabilizing effects. The wind continuously acts on exposed crane components and suspended payloads, generating aerodynamic forces that vary with wind speed, direction, turbulence intensity, and geometry. Extended booms and large payloads significantly increase wind-exposed surface area, while flexible hoisting systems allow payload swing and rotation. Accident investigations have shown that wind-related effects, including unexpected gusts and sustained crosswinds, have contributed to several crane tip-over incidents even when operations nominally complied with the limits of the load chart [3–5].

To mitigate these risks, industry practice relies primarily on manufacturer-provided load charts and standardized wind-load assessment procedures defined in documents such as ASME B30.5 and ISO 4302 [1, 6]. These standards provide baseline guidance and incorporate safety factors to account for uncertainties in loading and operating conditions. However, they are derived under quasi-static assumptions and typically treat the payload as a rigid, undeformed body subjected to simplified wind forces. As a result, deformation-induced effects—such as wind-driven payload deflection, cable elongation, and shifts in aerodynamic center of pressure—are not explicitly represented. As mobile cranes are increasingly deployed in wind-exposed and space-constrained environments, these limitations motivate the need for physics-based analysis methods capable of resolving the coupled interaction between wind flow, structural deformation, and crane stability. This motivation provides the context for the detailed examination of crane stability mechanisms and wind-induced tip-over risk presented here.

In classical crane stability analysis, the tipping condition is expressed as a balance between overturning and restoring moments about the outrigger or wheel edge acting as the pivot. Overturning moments arise from payload weight, boom weight, and externally applied loads such as wind, while restoring moments are generated by gravity acting through the crane's components and counterweights. This formulation implicitly assumes static loading and treats the suspended payload as a rigid, undeformed body [7]. Adams and Singhose have analyzed how stability margins related to configuration parameters like boom angle have become quite small as cranes have gotten taller and taller [8]. These smaller stability margins may not be sufficient to prevent tip-over in the presence of uncertainties and/or larger-than-expected external disturbances such as wind.

Several studies have demonstrated that dynamic effects can significantly influence crane stability. Eden et al. showed that inertial forces arising from crane operations such as slewing, hoisting, and braking can reduce stability margins [9]. Building on this, Zaretsky and Shapiro proposed an energy-based overturning criterion, where tip-over occurs when the kinetic energy introduced during dynamic events exceeds the restoring potential energy about the tipping axis [10]. Further work by Rauch and Fujioka incorporated payload motion and showed that even moderate payload swing can substantially reduce allowable load capacity [11, 12]. While these studies highlight the importance of dynamic and inertial effects, they primarily address crane motion-induced loading. In contrast, the present work focuses on mean wind loading as the dominant source of aerodynamic forces and overturning moments, and evaluates stability under quasi-static conditions.

Aerodynamic forces depend on flow direction and geometry, and act continuously over time. In engineering practice, wind effects are often represented as equivalent static horizontal forces applied at the payload or boom tip using empirical drag coefficients [1, 5]. Such simplifications neglect three-dimensional flow separation, wake interaction between crane components and payloads, and deformation of suspended loads, and therefore fail to capture changes in projected area, center-of-pressure location, and effective moment arm caused by wind-induced structural deflections and payload swing.

Although computational fluid dynamics (CFD) has been used to estimate wind loading on crane booms and large construction equipment [13, 14], the combined influence of wind-induced deformation on mobile crane tip-over stability remains poorly addressed.

In the present work, the focus is on mean aerodynamic loading and the resulting quasi-static structural response. While wind loading in reality contains fluctuating components, the objective here is to quantify how mean wind forces interact with structural deformation to influence overturning moments and stability.

In the present framework, this interaction is modeled using a one-way CFD-to-structural coupling, where deformation-induced changes in effective moment arm are captured, while feedback to the flow field is neglected. Dynamic effects such as resonance, damping-controlled response, and turbulence-induced amplification are not modeled in this work and are considered as part of future extensions. Classical FSI theory emphasizes that accurate prediction of such systems requires consistent treatment of aerodynamic forces and structural response [15]. While FSI methods have been extensively developed and applied in aeroelasticity and flexible-structure dynamics, crane-related studies have historically focused on inertial excitation and payload sway control rather than wind-induced deformation [15, 16].

The remainder of this paper presents a detailed numerical investigation of wind-induced stability of a mobile crane using a one-way CFD–structural framework under mean wind loading. The computational modeling approach, including geometric representation of the crane–payload system, governing equations, and numerical implementation of the fluid and structural solvers, is described in Section 2. The aerodynamic and structural response results are then examined in Section 3 to quantify wind-induced forces, payload deflection, and the resulting overturning moments for different wind directions and payload configurations. These results are compared with conventional rigid-body and standards-based stability estimates in Section 4 to highlight the influence of deformation-dependent effects. Finally, the key findings are summarized, and the implications for crane stability assessment are discussed along with limitations of the present study and directions for future work.

2. Methodology and Numerical Setup

2.1. Overall Computational Framework

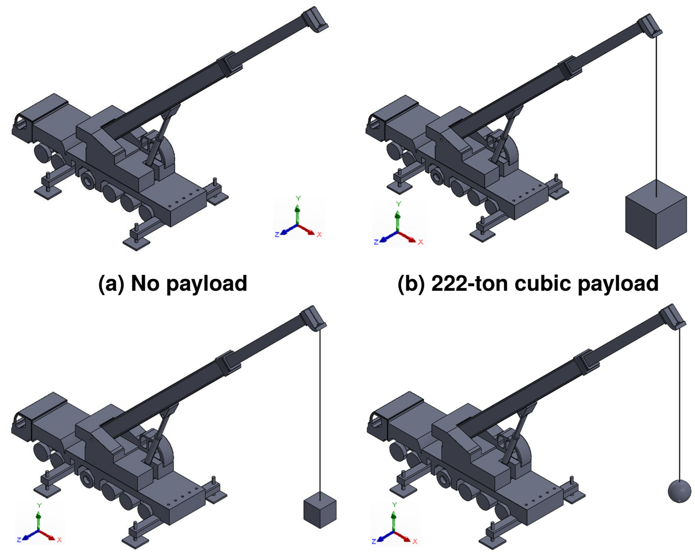
The present study adopts a sequential one-way CFD–structural (FSI-inspired) methodology to evaluate mean wind-induced aerodynamic loading and its influence on stability through quasi-static structural response of a mobile crane with suspended payloads. The workflow consists of three main stages: (i) geometric modeling of the crane–payload system, (ii) wind-flow simulation using ANSYS Fluent to compute aerodynamic forces and moments, and (iii) structural response analysis using ANSYS Static Structural with imported CFD loads to quantify payload deflection and deformation-dependent overturning moments. All simulations were performed using ANSYS 2025 R1 Fluent and Mechanical. The dynamic characteristics of telescopic crane systems typically involve low-frequency modes associated with boom flexibility and payload motion. However, for the present configurations, the aerodynamic loading is dominated by mean wind effects, and transient CFD simulations indicate negligible variation in drag and moment compared to steady-state results. Therefore, dynamic amplification effects are not expected to significantly influence the stability response for the cases considered.

In the FSI workflow fluid domain, ANSYS Fluent was used to obtain converged surface pressure and shear stress distributions on the crane and payload. These aerodynamic loads were then transferred unidirectionally to the Static Structural solver, where they were applied as external loads to compute structural deformation of the boom–rope–payload system. This assumption is appropriate for the present phase of the study, as payload deflections remain small relative to the overall crane geometry and do not significantly alter the global flow topology. The computed deflections are around 1 m for the studied cases, which is small relative to characteristic crane dimensions. Therefore, deformation-induced feedback to the flow field is neglected, and a one-way coupling approach is adopted.

2.2. Geometric Modeling and Crane Configuration

A detailed three-dimensional model of a 175-ton all-terrain telescopic truck crane was developed using SolidWorks, inspired from Linkbelt's 175|AT cranes [17]. The boom is fixed at a 45° elevation angle and oriented toward the rear of the vehicle, a configuration identified in industry practice and load charts as one of the least stable operating conditions.

The payload configurations shown in Figure 2 are investigated to evaluate the influence of mass and geometry on aerodynamic loading and stability.



(a) No payload **(b) 222-ton cubic payload**

(c) 27-ton cubic payload **(d) Spherical payload**

FIGURE 2: Crane configurations used in the numerical simulations for evaluating the effects of payload mass and geometry on aerodynamic loading and quasi-static stability.

TABLE 1: Summary of Computational Mesh Parameters

Parameter	Value / Setting
Meshing tool	ANSYS Meshing
Solver preference	ANSYS Fluent
Physics preference	CFD
Mesh type	Unstructured (hybrid)
Total nodes	5,025,319
Total elements	27,830,601
Enclosure (fluid domain) sizing	0.5 m (body sizing)
Crane & payload surface sizing	0.03 m (face sizing, 675 faces)

The hoist rope is explicitly modeled in the structural analysis to capture payload deflection under combined gravitational and wind loading.

2.3. Mesh Generation and Grid Statistics

The computational mesh was generated using ANSYS Meshing with solver preference set to ANSYS Fluent. A hybrid unstructured mesh was employed, with localized refinement applied near the crane and payload surfaces and a coarser mesh used in the far-field enclosure. Curvature-based refinement was enabled to accurately capture geometric features, while proximity-based refinement was not applied.

A body sizing control was applied to the enclosure to regulate far-field resolution and reduce computational cost, while face sizing controls were applied to crane and payload wall faces to accurately resolve surface pressure distributions required for aerodynamic force integration. The resulting mesh provided sufficient resolution to capture flow separation, wake interaction, and pressure loading on the crane–payload system, while maintaining numerical stability and computational efficiency for the CFD–FSI analysis. Table 1 summarizes the key meshing choices and grid statistics used in the CFD simulations.

Mesh resolution was refined near the crane surfaces to capture boundary-layer effects. Figure 3 illustrates how the enclosure

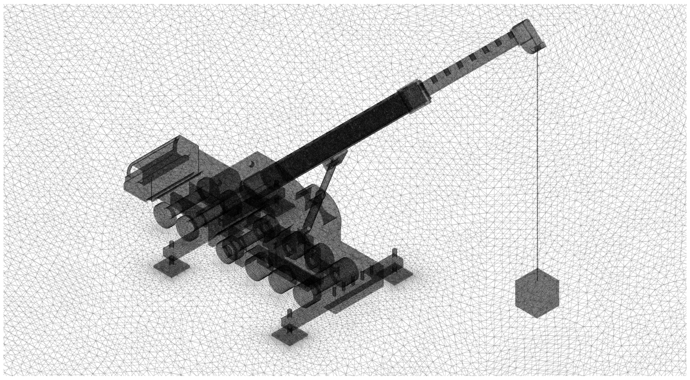


FIGURE 3: Body sizing and Face sizing contrast in meshing

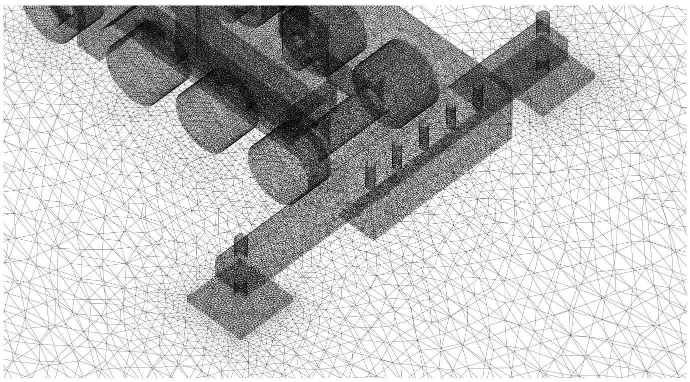


FIGURE 4: Mesh size transition between enclosure and crane wall

body has larger mesh size compared to the crane wall in order to preserve computational efficiency. The gradual transition in mesh size from enclosure body to crane wall at the outrigger is demonstrated in Figure 4 as an example.

2.4. Mesh Convergence Study

To ensure that the computed aerodynamic loads are independent of grid resolution, a mesh convergence study was performed for a representative crane configuration under crosswind loading. Three levels of mesh refinement were considered: a coarse mesh, a medium mesh, and a fine mesh, with total cell counts of approximately 0.37, 6.7, and 27 million, respectively.

All meshes were generated using the same meshing strategy described in Section 2.3, with consistent boundary conditions and solver settings. Local refinement near crane and payload surfaces was preserved across all mesh levels, while the global mesh density was varied to assess sensitivity to resolution.

The primary quantities used for convergence assessment were the integrated aerodynamic drag force and the resulting overturning moment about the tipping axis. These quantities are directly relevant to the stability analysis and therefore serve as appropriate metrics for evaluating mesh independence.

The results of the mesh convergence study are summarized in Table 2. The variation in drag force between the medium and fine meshes was found to be [3.4%], while the corresponding variation in overturning moment was [3%]. These differences are within acceptable tolerance limits for engineering analysis, indicating that the solution is sufficiently grid-independent.

Based on these results, the finer mesh was selected for all subsequent simulations, as it provides a suitable balance between computational efficiency and solution accuracy. The chosen mesh

TABLE 2: Mesh convergence study for aerodynamic drag and overturning moment

Mesh	Cells	Drag (N)	Moment (Nm)	% Difference
Coarse	[376871]	[5477]	[36238]	–
Medium	[6717430]	[5597]	[38891]	[7%]
Fine	[27830601]	[5794]	[40078]	[3%]

resolution is therefore considered adequate for capturing the relevant flow features, including pressure distribution, wake development, and aerodynamic loading on the crane–payload system.

2.5. Governing Equations and Numerical Methods

Fluid Flow Governing Equations

The instantaneous wind velocity is expressed as

$$\mathbf{U}(t) = \overline{\mathbf{U}} + \mathbf{u}'(t), \tag{1}$$

where $\overline{\mathbf{U}}$ is the mean wind speed and $\mathbf{u}'(t)$ represents fluctuations [18]. In the present study, only the mean component is considered in the aerodynamic loading analysis.

To facilitate practical design, most wind-loading standards adopt equivalent-static formulations. A commonly used representation is the classical drag formulation,

$$F_D = \frac{1}{2}\rho C_D A U^2, \qquad M_{\text{wind}} = F_D h_{\text{cp}}, \tag{2}$$

where ρ is the air density, C_D is the drag coefficient, A is the projected area, U is the mean wind speed, and h_{cp} is the assumed height of the aerodynamic center of pressure. Projected area and center of pressure are assumed fixed.

Wind flow around the crane–payload system is modeled as an incompressible viscous fluid governed by the Reynolds-Averaged Navier–Stokes equations (RANS) [19]. The conservation of mass and momentum are expressed as

$$\nabla \cdot \mathbf{u} = 0, \tag{3}$$

$$\rho\left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u}\cdot\nabla\mathbf{u}\right) = -\nabla p + \nabla\cdot\left[(\mu+\mu_t)(\nabla\mathbf{u} + \nabla\mathbf{u}^T)\right], \tag{4}$$

where $\mathbf{u}$ is the velocity vector, p is the pressure, ρ is the air density, μ is the molecular viscosity and μ_t is the turbulent eddy viscosity. Turbulence effects are modeled using the shear stress transport (SST) k–ω model [20], which provides improved prediction of flow separation and wake development around bluff bodies such as crane booms and payloads. The transport equations for the turbulence kinetic energy k and the specific dissipation rate ω are solved concurrently to determine μ_t. This turbulence closure is well suited for wind loading problems involving adverse pressure gradients and separation. In the present study, steady-state RANS solutions are used to obtain mean aerodynamic loads.

Aerodynamic loads acting on the crane and payload are obtained by integrating pressure and viscous stresses over the wetted surfaces [21],

$$\mathbf{F}_{\text{aero}} = \int_A (-p\mathbf{n} + \boldsymbol{\tau}\cdot\mathbf{n})\, \mathrm{d}A, \tag{5}$$

where $\mathbf{n}$ is the outward surface normal and $\boldsymbol{\tau}$ is the viscous stress tensor. These forces are resolved into global coordinate directions to compute drag, lift, and moment components.

Structural Mechanics Governing Equations

The structural response of the crane–payload system is modeled using linear elasticity under quasi-static conditions [22]. The governing equation for structural equilibrium is

$$\nabla \cdot \boldsymbol{\sigma} + \mathbf{f}_b = 0, \tag{6}$$

where $\boldsymbol{\sigma}$ is the Cauchy stress tensor and $\mathbf{f}_b$ represents body forces, including gravity. The stress–strain relationship is given by Hooke's law,

$$\boldsymbol{\sigma} = \mathbf{C} : \boldsymbol{\varepsilon}, \tag{7}$$

where $\mathbf{C}$ is the elastic stiffness tensor and $\boldsymbol{\varepsilon}$ is the strain tensor derived from displacement gradients. Material properties corresponding to structural steel are used for the boom, payload, and rope elements.

Appropriate boundary conditions were applied and hoist rope was modeled to capture axial deformation, enabling realistic prediction of payload deflection under combined gravitational and aerodynamic loads.

Fluid–Structure Interaction Coupling

The interaction between aerodynamic loading and structural response was modeled using a one-way FSI approach. In this framework, the fluid and structural problems were solved sequentially, with aerodynamic loads computed from CFD applied as external loads in the structural solver.

At the fluid–structure interface, traction continuity is enforced in the form

$$\mathbf{t}_s = \mathbf{t}_f = (-p\mathbf{n} + \boldsymbol{\tau} \cdot \mathbf{n}), \tag{8}$$

where $\mathbf{t}_f$ and $\mathbf{t}_s$ denote the fluid and structural tractions, respectively. Displacement compatibility was not enforced in the fluid domain due to the one-way coupling assumption [23].

This approach was justified because the magnitude of payload deflection was small relative to the characteristic dimensions of the crane and did not significantly modify the global flow field. However, the resulting deformation was sufficient to alter the effective moment arm and center of pressure, which directly influenced overturning moments.

Evaluation of Overturning Moments and Aerodynamic Load

Surface pressure and shear stress distributions obtained from CFD simulations are integrated over individual crane components and payload surfaces to compute aerodynamic forces. These forces are resolved into the global coordinate system and used to evaluate moments about the crane base and tipping axes.

The wind-induced overturning moment about a tipping axis is evaluated as a deformation-dependent quantity,

$$\mathbf{M}_{\text{overturning}} = \int \mathbf{r}(t) \times \mathbf{F}_{\text{aero}}(t)\, \mathrm{d}t, \tag{9}$$

where $\mathbf{F}_{\text{aero}}(t)$ is obtained from CFD and $\mathbf{r}(t)$ represents the position vector from the tipping axis to the instantaneous line of action of the aerodynamic force, updated based on structural deformation [10, 24, 25].

This formulation captures the amplification of overturning moments caused by wind-induced payload deflection and rope elongation—effects neglected in conventional rigid-body stability analyses.

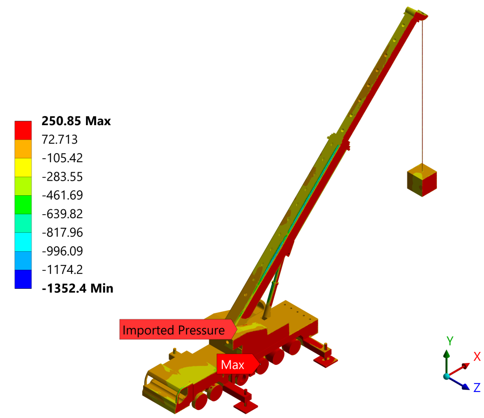


FIGURE 5: Imported pressure for crosswind

The study evaluates aerodynamic loading for multiple wind speeds, enabling assessment of force and moment scaling with velocity. The pressure distribution for the crosswind loading case is shown in Figure 5, and results are tabulated to compare drag forces, lift forces, and overturning moments across different wind directions and payload configurations.

2.6. CFD Domain and Numerical Setup

The computational domain consists of a rectangular enclosure sized to minimize blockage effects and allow adequate development of the wake downstream of the crane. Wind was imposed using a velocity inlet boundary condition, with two primary wind orientations considered:

- 0° wind direction, aligned toward the rear of the crane, and
- 90° wind direction, representing lateral crosswind loading.

The enclosure was created around the crane with inlet and outlet walls defined based on wind direction for aligned and crosswind cases. The ground plane was modeled as a no-slip wall, while lateral and top boundaries were treated as slip or symmetry boundaries. Air was modeled as an incompressible fluid. A steady-state Reynolds-Averaged Navier–Stokes (RANS) formulation was employed using the SST k–ω turbulence model to resolve separation and wake effects around the boom and payload.

In addition, transient (unsteady RANS) simulations were performed for selected representative cases to assess the influence of time-dependent effects on aerodynamic loading. These simulations were conducted over a total duration of 10 s with a time step of 0.02 s. Comparison of transient and steady-state results showed negligible differences in integrated aerodynamic quantities such as drag force and overturning moment, indicating that mean aerodynamic loading dominates for the configurations considered. Based on this observation, steady-state RANS solutions were adopted for all parametric studies, and the resulting aerodynamic loads were used in a quasi-static structural analysis. The variation between transient and steady-state predictions of drag and overturning moment was within 5%, confirming that dynamic fluctuations do not significantly influence the global loading for the studied cases.

2.7. Structural Analysis and Fluid–Structure Interaction

Fixed boundary conditions were applied at the outrigger plate bases, representing full ground contact. Gravitational acceleration was applied in the vertical direction, and CFD-derived pressure loads were imported directly onto the crane and payload surfaces. The structural response is evaluated under quasi-static loading conditions, neglecting inertial and damping effects.

The structural analysis computed payload deflection, rope elongation, and deformation of the crane structure under combined gravitational and wind-induced loading. These deformations modify the effective moment arm of aerodynamic forces and are used to evaluate deformation-dependent moments.

3. Results

3.1. Effect of Payload Weight and Geometry on Aerodynamic Forces and Moments

Aerodynamic forces and moments at a wind speed of 18 m/s were obtained from steady-state CFD simulations for the multiple payload configurations and wind directions detailed in Section 2. Simulations were advanced to 750 iterations, achieving continuity residuals on the order of 10^{-3}, which was deemed sufficient for convergence of integrated force and moment quantities. Wind aligned with the crane body (0°) corresponds to inflow in the $-X$ direction, while crosswind loading (90°) corresponds to inflow in the $+Z$ direction seen in Fig 5. Moments were reported about the global origin $(0\,\mathrm{m}, 0\,\mathrm{m}, 0)$, with M_x, M_y, and M_z representing rotations about the respective coordinate axes. The forces and moments obtained from fluid simulations without consideration of the crane and payload weight are displayed in Table 3, with negative values indicating forces in the negative directions and moments in the counterclockwise directions.

For the 0° wind direction, the aerodynamic drag force increased from approximately 5.96 kN in the no-payload case to 6.40 kN for the 27-ton cubic payload and 7.46 kN for the 222-ton cubic payload. This increase is primarily attributable to the growth in projected frontal area associated with the suspended payload. The spherical payload produced a lower drag force of approximately 5.79 kN, reflecting its smoother geometry and reduced pressure drag relative to the cubic payloads. Projected area of 27-ton cubic payload is 1.3 times the projected area of spherical payload of same size. These trends are consistent with classical drag scaling; however, the corresponding overturning moment component M_z exhibits comparatively smaller variation, ranging from -40.1 kN·m for the spherical payload to -46.1 kN·m for the 222-ton payload.

The aerodynamic drag force increases significantly with payload size and projected area. For the 0° wind direction, drag increases from 5.96 kN for the no-payload case to 22.64 kN for the 27-ton cubic payload, primarily due to the larger exposed area. The spherical payload exhibits lower drag compared to cubic configurations due to its smoother geometry and reduced flow separation as demonstrated in Figure 8.

Crosswind loading (90°) produces substantially higher forces and moments compared to along-wind conditions. This is attributed to increased projected area and altered flow interaction with the boom–payload system, resulting in stronger aerodynamic loading and moment amplification.

Despite the variation in force magnitudes, the overturning moment is not solely governed by drag. Instead, it is strongly influenced by the effective moment arm, which depends on the position of the aerodynamic center and deformation of the crane–payload system. For the along-wind case, the moment arm is largely controlled by payload height, whereas for crosswind loading, both payload position and boom deformation contribute significantly.

Overall, the results demonstrate that both payload geometry and orientation play a critical role in determining aerodynamic loading and stability. Cubic payloads produce higher forces and moments due to larger projected areas, while spherical payloads result in reduced aerodynamic loading. Crosswind conditions consistently represent the most critical loading scenario.

3.2. Effect of Wind Velocity and Wind Direction on Aerodynamic Loading and Overturning Moments

The influence of wind velocity and wind direction on aerodynamic loading and overturning moments was examined using the spherical payload configuration, which provides a clean baseline due to its smooth geometry and reduced sensitivity to flow separation. Results are presented for four wind velocities (7.07, 10.62, 14.14, and 18 m/s) under both along-body (0°) and crosswind (90°) orientations.

For along-body wind loading (0°), aerodynamic forces and moments increased monotonically with wind velocity, consistent with the quadratic dependence of drag on velocity. As displayed in Table 4, the drag force in the wind-aligned direction increased from approximately 0.9 kN at 7.07 m/s to nearly 5.8 kN at 18 m/s, while the magnitude of the corresponding overturning moment about the principal tipping axis increased by approximately 33 kN·m. Although the payload contributed directly to this increase, the magnitude of the moment remained moderated by partial aerodynamic shielding from the boom and the relatively small projected area of the spherical payload in the wind direction. As a result, the growth in overturning moment with velocity remained smooth and predictable.

In contrast, crosswind loading (90°) produced substantially larger aerodynamic moments at all wind velocities. At 18 m/s, the overturning moment magnitude exceeded 75 kN·m, nearly doubling the along-wind value despite comparable drag magnitudes. This amplification arises primarily from geometric effects rather than force magnitude alone. Under crosswind conditions, the payload and boom present a larger projected area, and the aerodynamic center of pressure shifts laterally, increasing the effective moment arm relative to the tipping axis. Consequently, even moderate increases in aerodynamic force translate into significant increases in overturning moment.

The velocity dependence under crosswind conditions further highlights this coupling. As wind speed increases, both drag scaling and enhanced flow separation around the payload–rope system contribute to moment amplification. The moment did not scale linearly with drag, indicating that geometric leverage and load redistribution dominate the stability response. Across all velocities, the spherical payload exhibited lower drag and moment values than cubic payloads at comparable wind speeds, confirming that smoother geometry promotes attached flow and reduces pressure drag. However, crosswind loading remains the critical destabilizing condition even for this geometry, underscoring the importance of wind direction in crane stability assessment. These results demonstrate that wind velocity and direction are strongly coupled in determining crane stability.

TABLE 3: Aerodynamic forces and overturning moments for different payload configurations at a wind speed of 18 m/s under (a) 0° along-body wind and (b) 90° crosswind conditions.

Actual velocity	0° wind direction					90° wind direction				
	Drag force$_x$ (N)	Lift force (N)	Moment$_x$ (Nm)	Moment$_y$ (Nm)	Moment$_z$ (Nm)	Drag force$_z$) (N	Lift force (N)	Moment$_x$ (Nm)	Moment$_y$ (Nm)	Moment$_z$ (Nm)
18 m/s (222-ton payload)	7463	−544	−1857	3123	−46111	−18252	2026	−88717	102036	−11377
18 m/s (27-ton payload)	6402	−851	−1713	2377	−43073	−15792	2230	−79629	70173	−7316
18 m/s (No payload)	5959	−809	2604	−832	−43858	−14531	3437	−71623	56165	−3197
18 m/s (Spherical payload)	5794	−518	−1206	2099	−40078	-15780	4395	-76631	64467	−10375

TABLE 4: Aerodynamic forces and overturning moments for the spherical payload under along-body (0°) and crosswind (90°) loading at multiple wind velocities.

	0° Wind (Along-Body)					90° Wind (Crosswind)				
Velocity	$F_x(N)$	$F_y(N)$	$M_x(Nm)$	$M_y(Nm)$	$M_z(Nm)$	$F_z(N)$	$F_y(N)$	$M_x(Nm)$	$M_y(Nm)$	$M_z(Nm)$
18 m/s	5794	−518	−1206	2099	−40078	15780	4395	76631	−64467	−10375
18 m/s (sideward tipping)	–	–	–	–	–	–	–	121954	−64506	−10390
18 m/s (backward tipping)	–	–	−1559	658	−45994	–	–	–	–	–
14.14 m/s	3531	−228	−635	1101	−24300	9745	2721	47299	−39780	−7031
14.14 m/s (sideward tipping)	–	–	–	–	–	–	–	75320	−39771	−7028
14.14 m/s (backward tipping)	–	–	−799	434	−28537	–	–	–	–	–
10.62 m/s	2005	−155	−281	585	−13764	5548	1393	26881	−22587	−4370
10.62 m/s (sideward tipping)	–	–	–	–	–	–	–	42137	−22747	−4429
10.62 m/s (backward tipping)	–	–	−435	−43	−16003	–	–	–	–	–
7.07 m/s	886	−62	−275	393	−6233	2450.7	627	11923	−9999	−1858
7.07 m/s (sideward tipping)	–	–	–	–	–	–	–	18715	−10016	−1864
7.07 m/s (backward tipping)	–	–	−354	72	−7266	–	–	–	–	–

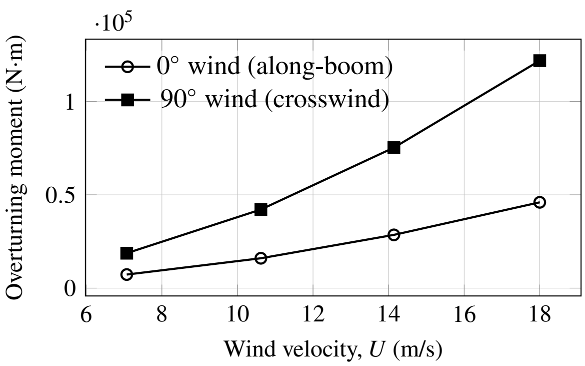


FIGURE 6: CFD-predicted overturning moments about the candidate tipping edges as a function of wind velocity for along-boom (0°) and crosswind (90°) loading.

3.3. Moments About Candidate Tipping Axes and Implications for Overturning

While moments about the origin are useful for reporting overall loading trends, they may understate or misrepresent tipping tendency if the origin is not located at or near the tipping axis. In reality, crane stability is governed by rotation about a support edge rather than about an arbitrary reference point. Accordingly, overturning moments were evaluated about physically meaningful tipping points identified from outrigger geometry.

Figure 6 displays a comparison between tipping moments for parallel and crosswind cases with respect to wind velocities. It is observed that crosswind causes more overturning moment and that it increases with velocity.

Direction-dependent amplification when shifting the reference point

For the spherical payload, shifting the moment reference from the origin to the tipping axis produced a clear increase in the magnitude of the relevant overturning components. This follows directly from rigid-body mechanics, where the moment about a point P is given by

$$\mathbf{M}_P = \mathbf{M}_O - \mathbf{r}_{OP} \times \mathbf{F}. \qquad (10)$$

where $\mathbf{M}_P$ is the moment vector about point P, $\mathbf{M}_O$ is the moment vector about the reference origin O, $\mathbf{r}_{OP}$ is the position vector from O to P, and $\mathbf{F}$ is the resultant aerodynamic force vector.

When the reference point is moved toward the tipping edge, the effective lever arm associated with aerodynamic forces changes. Under crosswind loading (90°), this effect was particularly strong because the aerodynamic force had a large lateral component and the offset to the sideward tipping point increased the moment arm relative to the pivot. For instance, at 18 m/s the moment about the origin was 7.66×10^4 N·m, whereas the moment about the tipping point increases to approximately 1.22×10^5 N·m. Similar amplification was observed at velocities of 14.14 m/s, 10.62 m/s and 7.07 m/s. This demonstrates that the configuration is closer to tip-over when assessed about the true pivot and reinforces that stability should be evaluated about the tipping edge rather than about the crane's geometric center or model origin.

For 0° wind loading, moments about the candidate tipping axes also differed from origin-based moments, but the amplification was generally smaller than in crosswind cases. This aligns with the earlier observation that along-wind loading is more strongly influenced by drag acting on the crane superstructure and that the payload is partially shielded by the boom. Consequently, the net force direction and line of action shifted less dramatically with reference-point changes, and the tipping-point moment increased more moderately. Nonetheless, tipping-point values remain the correct basis for overturning evaluation because they directly represent rotation tendency about the support edge.

Velocity scaling and tipping sensitivity

Across the velocity sweep, tipping-point moments increased strongly with wind speed, consistent with the quadratic scaling of aerodynamic loads with velocity. However, the key result is that tipping-point moments grew faster in a practical sense than origin-based moments because the tipping-point formulation incorporated the correct lever arm. This effect is particularly evident under 90° loading, where increases in projected area and lateral shifts of the aerodynamic center of pressure produced large destabilizing moments even at intermediate wind speeds (10.62–14.14 m/s). In other words, evaluating moments about the tipping edge reveals that crosswind conditions can approach critical overturning thresholds at wind speeds that may appear benign when using origin-based reporting or simplified static approximations.

The tipping-point moment results provide a direct bridge to the fluid–structure interaction analysis. In rigid-body CFD post-processing, the payload is assumed fixed in position, so the moment arm is determined purely by the geometry of the undeformed configuration. Wind loading induces rope inclination, payload rotation, and lateral displacement, which effectively increase the distance between the pivot and the aerodynamic resultant. This increases the overturning moment beyond the rigid-payload prediction by modifying $\mathbf{F}_{\text{aero}}$.

Accordingly, in the subsequent FSI results, overturning moment was evaluated using the same tipping-point reference and compared against the available restoring moment derived from crane weight distribution and counterweight–outrigger geometry.

Comparison of rigid-CFD tipping-point moments against deformation-informed FSI moments was used to quantify the additional reduction in stability margin attributable to wind-induced payload deflection—an effect not addressed in conventional standards-based evaluations.

3.4. Wind-Induced Structural Response and Deformation-Driven Overturning Effects

One-Way Fluid–Structure Interaction Results.

Wind-induced structural response was evaluated using a one-way FSI approach, in which converged CFD pressure loads were mapped onto the structural model of the boom–rope–payload assembly. The resulting deformation fields indicate that the dominant compliance occured in the hoist rope and suspended payload, producing measurable lateral displacement and rope inclination under wind loading. Structural deformation of the boom itself remained comparatively small.

Because aerodynamic loads were not recomputed on the deformed configuration, the present analysis does not include aerodynamic feedback effects (i.e., no two-way coupling). Nevertheless, the one-way FSI solution quantifies deformation-induced changes in payload position that directly influence stability by increasing the effective lever arm of aerodynamic forces about the candidate tipping edge. This mechanism is not represented in equivalent-static wind-load procedures, such as ISO 4302, which assumes rigid payload geometry and a fixed center of pressure.

Figure 7 demonstrates deflection of payload at 18 m/s obtained from one-way FSI. The tensile expansion of rope and deformation of payload is taken into consideration in the simulation, which becomes predominant at high speeds. At a wind speed of 7.07 m/s, the maximum total deformation extracted from the FSI solution was approximately 5.51 m in the along-wind view and 6.74 m in the crosswind-oriented view. These maxima occur in the rope–payload region, consistent with observed rope inclination and lateral payload shift.

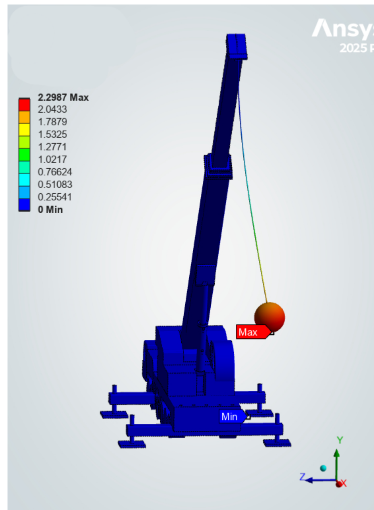


FIGURE 7: Payload swing for crosswind at 18 m/s

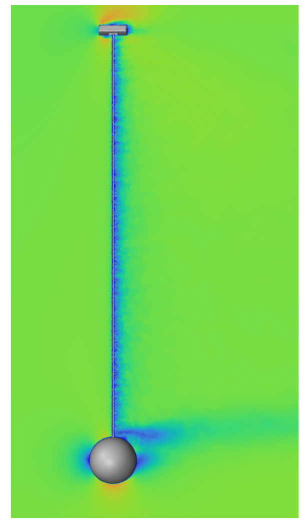


FIGURE 8: Velocity contour for crosswind

Structural response and Payload deflection. The structural response of the crane–payload system under combined gravitational and aerodynamic loading is plotted in Figure 9. The maximum payload deflection increases with wind velocity, consistent with the quadratic dependence of aerodynamic forces on wind speed. For the along-wind (0°) cases, deflection increases from 0.2498 m at 7.07 m/s to 1.3714 m at 18 m/s, while crosswind (90°) cases exhibit higher deflections due to increased projected area.

Despite this trend, the absolute magnitude of deflection remains small relative to the characteristic dimensions of the crane system (on the order of tens of meters). Even at the highest wind speed, the maximum deflection is approximately 1.37 m, indicating that structural deformation represents a limited geometric perturbation.

This observation supports the use of a one-way coupling approach, as the resulting deformation is not expected to significantly alter the global flow field or aerodynamic force distribution. At the same time, the deformation influences stability by modifying the effective moment arm of aerodynamic forces, thereby affecting overturning moments.

Overall, these results justify the adopted framework, in which steady aerodynamic loads are coupled with a quasi-static structural analysis to capture deformation-dependent stability effects

TABLE 5: Comparison of manually computed overturning moments vs ANSYS moments about sideward tipping axis for the spherical payload under crosswind loading.

Wind speed (m/s)	F_z (N)	F_y (N)	$M_{x,O}$ (N·m)	Manual $M_{x,P}$ (N·m)	ANSYS $M_{x,P}$ (N·m)	% diff
18.00	15779.59	4395.13	76630.70	121953.71	121953.70	~0.00001%
14.14	9744.81	2721.29	47298.54	75319.53	75319.53	~0.000002%
10.62	5548.12	1393.00	26880.50	42136.72	42136.90	~0.00044%
7.07	2436.82	673.73	11830.13	18807.00	18715.3	~0.48%

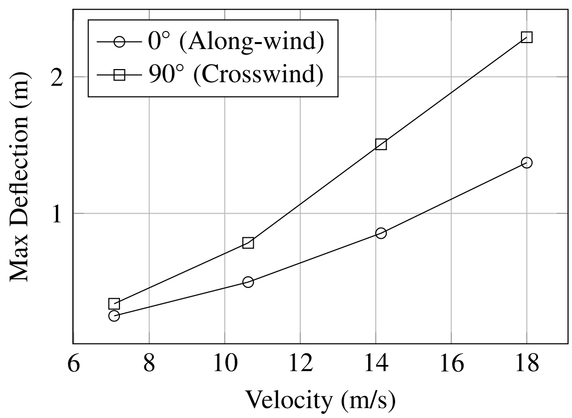


FIGURE 9: Variation of maximum payload deflection with wind velocity for along-wind and crosswind conditions

without requiring two-way fluid–structure interaction.

Geometry-Only Overturning Moment Amplification from One-Way FSI. To assess the stability implications of wind-induced deformation using the available one-way coupling results, a geometric amplification estimate was formed by combining the CFD-derived aerodynamic force with the FSI-predicted shift in payload position relative to the tipping axis. The incremental overturning contribution due to lever-arm change was approximated as

$$\Delta M_{\text{geom}} \approx F_{\perp}\, \Delta r_{\perp}, \tag{11}$$

where $F_{\perp}$ is the component of aerodynamic force producing rotation about the tipping axis and $\Delta r_{\perp}$ is the lateral displacement of the force line of action relative to the undeformed configuration. This estimate isolates deformation effects without invoking flow-field feedback and therefore provides a conservative indication of deformation-driven moment amplification.

For the 0° wind case at 7.07 m/s with a spherical payload, the CFD-derived drag force was $F_x = 886.2$ N. Using the maximum deformation magnitude $\Delta r \leq 0.2498$ m yields

$$\Delta M_{\text{geom}} \lesssim (886.2)(0.25) \approx 221.5 \text{ N} \cdot \text{m}. \tag{12}$$

This result indicates that even at modest wind speeds, deformation of the hoisting system can introduce a non-negligible additional overturning contribution that is not captured by rigid-body or equivalent-static analyses.

For crosswind loading (90°), the same procedure applies using the aerodynamic force component aligned with the crosswind direction, which was $F_z = 2450.7$ N and the corresponding deformation bound $\Delta r \leq 0.34$ m. Quantification of this contribution for 7.07 m/s yields

$$\Delta M_{\text{geom}} \lesssim (2450.7)(0.34) \approx 833.24 \text{ N} \cdot \text{m}. \tag{13}$$

To validate ANSYS "moment about a point" outputs, the moments about the tipping axes were computed manually from ANSYS-reported forces and moments using the rigid-body transformation. Table 5 displays moments about the tipping axes for all 4 velocities analyzed in the crosswind (90°) case. For 90° wind, overturning occurs primarily about the X-axis. As inferred from Table 5, for the spherical payload at 18 m/s, the manually computed value $M_{x,P} = 1.2195 \times 10^5$ N·m matches the ANSYS-reported value to numerical precision. Similar agreement was obtained for 14.14 m/s, 10.62 m/s and 7.07 m/s cases, validating the moment-shift procedure across wind speeds. For 0° wind, overturning is dominated by the moment about the Z-axis. For the spherical payload at 18 m/s, the manually computed tipping moment was -4.61×10^4 N·m, compared to -4.60×10^4 N·m reported directly by ANSYS about the same point. The discrepancy (< 0.2%) is attributable to numerical integration tolerance and rounding, confirming the correctness of the transformation.

Conclusions

This work presents a quasi-static CFD–FSI framework for evaluating wind-induced overturning effects in mobile cranes using one-way coupling. By transforming aerodynamic loads to the crane's actual tipping edges, the approach provides a configuration-specific measure of overturning moment consistent with the physical stability mechanism.

The results show that wind direction is the dominant driver of overturning response, with crosswinds producing significantly higher moments than along-wind loading across all payload configurations. Payload geometry also plays a critical role: cubic payloads generate larger aerodynamic forces due to higher projected area, while spherical payloads exhibit reduced drag and lower overturning moments. These findings highlight that payload shape, in addition to mass, must be considered in wind stability assessments.

The simulations further indicate that wind-induced deformation of the hoisting system contributes to increased overturning moments by modifying the effective lever arm of aerodynamic forces. This effect, captured through the CFD–FSI framework, is not represented in conventional equivalent-static methods such as ISO 4302, particularly under mean-wind loading conditions.

Additional transient simulations confirmed that time-dependent effects do not significantly influence aerodynamic forces and overturning moments for the configurations considered. Combined with the relatively small magnitude of structural deflections, this supports the use of a steady, one-way coupled framework for the present analysis.

Validation of the moment transformation procedure against

ANSYS results demonstrates the reliability of the proposed evaluation approach. While the present study focuses on deformation-dependent overturning effects, future extensions incorporating restoring moment estimation, two-way coupling, and gust response modeling can further enhance predictive capability and enable direct assessment of critical wind conditions.

Acknowledgments

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